

SEELAM THIRUMALA MURTHY BABU

Chennai, Tamil Nadu

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🌐 [seelam-thirumala-murthy-babu](https://seelam-thirumala-murthy-babu.github.io) 🌐 [SThirumalaMurthyBabu](https://SThirumalaMurthyBabu.github.io)

PROFILE

A hands-on Electrical Engineer who cares more about what's built than where it's studied. Driven by a genuine hunger to create, solve and bridge domains - grounded in EV powertrain systems and embedded protocols, now laser-focused on robotics, looking to plug into a team where that cross-domain thinking adds value from day one and grow into something the company counts on long-term.

TECHNICAL SKILLS

Languages: C, C++, Python, Shell Scripting
Robotics Frameworks: ROS2, Linux, CAN Protocol, Robot Programming, Turtlesim
Hardware & Electronics: Arduino, PCB Design (KiCAD), Proteus, PLC & SCADA
Engineering Tools: MATLAB, RViz, Gazebo, Git, GitHub, VSCode, Tinkercad, CANdb++
Soft Skills: Problem Solving, Time Management, Adaptability, Teamwork, Decision Making

WORK EXPERIENCE

Boson Motors Pvt Ltd

July 2025 - Present

Electrical Engineer - I

Chennai, Tamil Nadu

- Working across BVMC and Powertrain teams on the **LX40 - Boson's 4WD** electric utility vehicle - developing systems-level understanding of EV electrical architecture from vehicle parameter monitoring to subsystem fault diagnosis using **Grafana**, driving data-informed decisions across live vehicle data.
- Involved in end-to-end testing and validation of a next-generation powertrain developed in-house targeting improved efficiency and reduced manufacturing cost over the existing platform - contributing across component-level bench testing, test case preparation, data analysis, report generation and field testing.
- Developed hands-on working knowledge of **UART, I2C, SPI** and **CAN** protocols from concept to hardware-level implementation, extending into CAN network management by independently creating and maintaining DBC files for accurate signal decoding across vehicle subsystems - building the communication foundation directly transferable to robotic sensor and actuator integration.
- Took end-to-end responsibility for production validation of battery management, motor controller, DC-DC converter and on-board charging systems including CAN Isolator testing - ensuring subsystem readiness against defined acceptance criteria before vehicle-level integration.
- Developed **Python** scripts to automate CAN log and BMS log conversion into structured Excel graphs - cutting analysis time from 2-3 hours to under 5 minutes per log file, significantly accelerating debugging cycles regardless of log volume.
- Applied powertrain domain expertise beyond hardware boundaries - functioning as an informal powertrain-robotics integration thinker, bridging electrical architectures, embedded protocols and control signal analysis to mirror the sensing-to-actuation pipelines fundamental to autonomous robotic systems.

PROJECTS

Autonomous TurtleBot Navigation | ROS2, Turtlesim, Python, Linux

April 2026

- Built a TurtleBot simulation in ROS2 where a primary turtle autonomously navigates toward the nearest turtle, gaining practical understanding of nodes, topics and services, the core building blocks of robotic software systems.

Speed-Torque Mode Switching | CAN, BusMaster, VCU, Embedded C

Feb 2026

- Analysed and proposed two distinct approaches for **speed-torque mode switching** in the vehicle control system - a hardware-based digital input method through VCU code modification, and a software-based CAN switching method, evaluating trade-offs in wiring complexity, implementation effort and system reliability to identify the most efficient solution.

Drive-by-Wire System | CAN, BusMaster, DBC Files

Jan 2026

- Implemented electronic signal-based vehicle control over CAN bus by decoding DBC files and sending commands via BusMaster, directly mirroring how autonomous robotic systems decouple sensing from actuation.

CERTIFICATIONS

- ROS2 for Beginners (Jazzy) - Udemy
- Linux Essentials - Cisco Networking Academy
- HackerRank - Python 5 Star
- Wheeled Mobile Robots - NPTEL
- Embedded System Developer - Eduskills
- Cloud Security Essentials - AWS

ACHIEVEMENTS

- Received an international placement offer as Electrical Design Engineer at AEE, Tokyo, Japan during final-year B.Tech campus placements.
- First from my college to secure research internship offers at IIT Palakkad and IIT Tirupati during B.Tech.
- Minor in **CSE (CGPA: 9.3)** - completed in parallel to main B.Tech.
- Awarded Topper of the Year three times and **Sri Paul Prakash Memorial Award** in final year B.Tech.

EDUCATION

Course	College/University	Year	CGPA
BTech - Electrical & Electronics Engineering	Bapatla Engineering College	2022-25	9.68
Minors in Computer Science & Engineering	Bapatla Engineering College	2022-25	9.3
Diploma - Electrical & Electronics Engineering	Bapatla Polytechnic College	2019-22	9.11
High School	Little Angels High School	2019	9.7